

1 Basic things

Unit impulse | Delta

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}, \quad \delta_m[n] = \delta[n - m]$$

Unit step

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Exponential decay

$$x[n] = a^n u[n], \quad 0 < a < 1$$

Sinusoidal oscillation

$$x[n] = A \cos(\omega n + \phi), \quad A \in \mathbb{R}, \omega \in \mathbb{R}, \phi \in \mathbb{R}$$

Complex exponential

$$x[n] = A e^{j(\omega n + \phi)}, \quad A \in \mathbb{R}, \omega \in \mathbb{R}, \phi \in \mathbb{R}$$

Energy and Power

- $E_x = \|x\|^2$ (sum or integral of $|x|^2$)
- $P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$

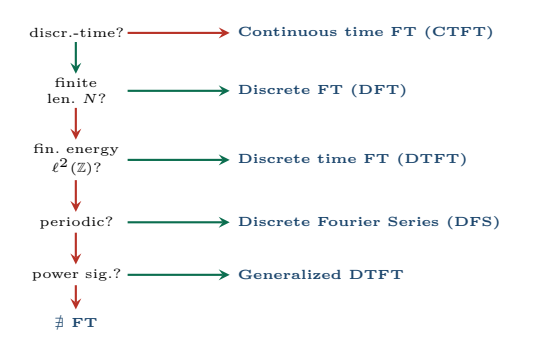
Operator combinations

	signal	sample at time n
	$\mathcal{R}S^d \mathbf{x}, \quad S^{-d} \mathcal{R} \mathbf{x}$	$x[-n + d]$
	$\mathcal{R}S^{-d} \mathbf{x}, \quad S^d \mathcal{R} \mathbf{x}$	$x[-n - d]$

Sum of a sequence

$$\sum_{k=m}^n k = \frac{(m+n)(n-m+1)}{2}$$
$$\sum_{k=m}^n z^k = \begin{cases} n - m + 1 & z = 1 \\ \frac{z^m - z^{n+1}}{1 - z} & z \neq 1 \end{cases}$$

2 Signal analysis



3 DFT: $X[k]$

Orthogonality (not orthonormality)

$$\langle \mathbf{w}_h, \mathbf{w}_k \rangle = N \delta[h - k], \text{ where}$$
$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n=0}^{N-1} x^*[n] y[n].$$

DFT matrix form

$$\mathbf{X} = \mathbf{W} \mathbf{x}, \quad \mathbf{x} = \frac{1}{N} \mathbf{W}^H \mathbf{X}, \text{ where } W_N = e^{-j \frac{2\pi}{N}}:$$
$$\mathbf{W}^H \mathbf{W} = N \mathbf{I}.$$
$$W_N^n = W_N^{n \bmod N} \text{ (aliasing property).}$$

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

4 Properties

A signal is **Hermitian-symmetric** if $x[n] = x^*[N - n]$.
DFS $\equiv$ DFT: same formulas, different interpretation (explicit periodicity).

5 DTFT: $X(e^{j\omega})$

Symmetry summary

Time domain $x[n]$	Freq. domain $X(\omega)$
Real	$X(-\omega) = X^*(\omega)$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

Generalized DTFT (Power Signals)
TODO [Explain the difference]

6 FFT

TODO [Explain the difference]

7 Spectrograms (STFT)

TODO [Explain the difference]

8 Filtering I - Digital Filters

LTI: Linearity & time-invariance

Let $y[n] = \mathcal{H}\{x[n]\}$.

Linearity ($\forall \alpha, \beta \in \mathbb{C}, \forall x_1, x_2$):

$$\mathcal{H}\{\alpha x_1[n] + \beta x_2[n]\} \stackrel{?}{=} \alpha \mathcal{H}\{x_1[n]\} + \beta \mathcal{H}\{x_2[n]\}$$

Time-Invariance ($\forall n_0 \in \mathbb{Z}, \forall x$):

$$\underbrace{\mathcal{H}\{x[n - n_0]\}}_{x[\cdot] \leftarrow x[\cdot - n_0]} \stackrel{?}{=} \underbrace{y[n - n_0]}_{n \leftarrow n - n_0}$$

Impulse response

$\mathbf{h} = \mathcal{H}\delta$. An LTI system is *fully* described by $\mathbf{h}$.

Convolution $\mathbf{y} = \mathbf{x} * \mathbf{h}$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Convolution properties

Commutativity: $\mathbf{x} * \mathbf{h} = \mathbf{h} * \mathbf{x}$

Linearity: $\mathbf{x} * (\alpha \mathbf{y} + \beta \mathbf{w}) = \alpha(\mathbf{x} * \mathbf{y}) + \beta(\mathbf{x} * \mathbf{w})$

Time-shift: $\mathbf{x} * (\mathcal{S}^k \mathbf{y}) = \mathcal{S}^k(\mathbf{x} * \mathbf{y})$

Associativity: $(\mathbf{x} * \mathbf{h}) * \mathbf{w} = \mathbf{x} * (\mathbf{h} * \mathbf{w})$

Cascade: $\mathbf{h} = \mathbf{h}_1 * \mathbf{h}_2$; order is irrelevant.

FIR vs IIR (Finite/Infinite Impulse Response)

- **FIR:** $\mathbf{h}$ finite support. No feedback;
- **IIR:** $\mathbf{h}$ infinite support.

Definition of causal

Only responds to present or past inputs.

BIBO Stability

LTI is BIBO stable $\iff \mathbf{h}$ is absolutely summable

Frequency Response

We extract from it the magnitude/phase response.

$$H(\omega) = \text{DTFT}\{\mathbf{h}[n]\}$$

Filter types by magnitude response

- **Lowpass:** passes low freqs ($\omega \approx 0$)
 - **Highpass:** passes high freqs
 - **Bandpass:** passes freqs near $\pm\omega_c$
 - **Allpass:** $|H(\omega)| = \text{const}$ — modifies phase only
- Stop Band: 0 - Transition Band - Pass Band: 1.

Unit-circle modulus reflection identity

$$\left|1 - az^{-1}\right| = |a| \cdot \left|1 - a^{-1}z^{-1}\right|$$

Example of filters

Moving Average (MA)

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k]$$

$h[n] = \frac{1}{M}$ for $0 \leq n < M$, else 0. Causal FIR $\Rightarrow$ always BIBO stable. Big cost.

Leaky Integrator (LI)

$$y[n] = \lambda y[n - 1] + (1 - \lambda)x[n], \quad |\lambda| < 1$$

$h[n] = (1 - \lambda)\lambda^n u[n]$ (causal, exponential decay)

Causal IIR. Constant cost. Approximates MA of length M with $\lambda = \frac{M-1}{M}$.

9 Filtering II - Filter Design

Transfer function $H(z)$

$$H(z) = \frac{B(z)}{A(z)} = b_0 \frac{\prod_{k=0}^{M-1} (1 - z_k z^{-1})}{\prod_{k=0}^{N-1} (1 - p_k z^{-1})}$$

$B(z)$: feedforward (FIR part).

$A(z)$: feedback (IIR part). FIR: $A(z) = 1$.

Zeros z_k : roots of $B(z)$, where $|H| = 0$.

Poles p_k : roots of $A(z)$, where $|H| \rightarrow \infty$.

Zero symmetry

If $h[n]$ is **real**: zeros in conjugate pairs.

If $h[n]$ is **symmetric** (linear phase): zeros in reciprocal pairs ($z_0, 1/z_0^*$).

ROC & causality

$$z \in \text{ROC}\{X(z)\} \iff \sum_{n=-\infty}^{\infty} |x[n]z^{-n}| < \infty$$

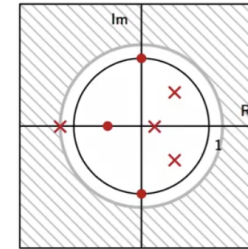
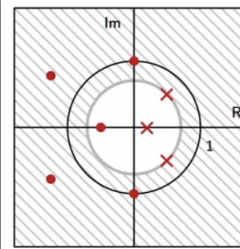
Causal IIR: $\text{ROC} = \{|z| > R\}$: $R = \max_k |p_k|$.

FIR: $\text{ROC} = \mathbb{C}$.

BIBO stability

Stable $\iff$ unit circle $|z| = 1$ lies inside the ROC
 $\iff$ all poles strictly inside the unit circle ($|p_k| < 1$).

Stable (unit circle in ROC) Unstable (unit circle not in ROC)



10 Filtering III - Stochastic and Adaptive SP

Reminder - Proba

CDF: $F_X(\alpha) = P(X \leq \alpha)$

PDF: $f_X(\alpha) = \frac{d}{d\alpha} F_X(\alpha)$

Moments: $E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$

Variance: $\sigma_x^2 = E[(x - \mu_x)^2]$

Cross-correlation & Covariance

Cross-correlation: $R_{xy} = E[xy]$

Covariance: $C_{xy} = E[(x - m_x)(y - m_y)]$

Uncorrelated: $E[xy] = E[x]E[y] = m_x m_y$

Independent: $f_{xy}(x, y) = f_x(x)f_y(y)$

Cross-Correlation

$$r_{xy}[k] = \langle \mathbf{x}, \mathcal{S}^k \mathbf{y} \rangle = \sum_{n=-\infty}^{\infty} x^*[n] y[n + k]$$

White Noise Process

Zero-mean: $E[x[n]] = 0$

Uncorr.: $E[x[n]x[m]] = E[x[n]]E[x[m]]$ for $m \neq n$

Autocorrelation: $r_x[n] = \sigma_x^2 \delta[n]$

PSD: $P_w(\omega) = \sigma_w^2$

Wide-Sense (WSS) stationarity

For WSS random processes we only care about the first two moments:

- $E[x[n]] = m_x$
- $E[x[n]x[m]] = r[n - m]$

Autocorrelation (not given)

$$x[n] = a^n u[n] \Rightarrow r_x[k] = \frac{a^{|k|}}{1 - a^2}$$

$$\text{DTFT}\{r_x[k]\} = \frac{1}{1 + a^2 - 2a \cos \omega}$$

Power Spectral Density (PSD)

Estimator (DFT): $P[k] = E[|X_N[k]|^2/N]$

Exact definition (Wiener-Khinchin):

$$P_x(\omega) = |X(\omega)|^2 = \text{DTFT}\{r_x[k]\}$$

Variance: $\sigma_x^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_x(\omega) d\omega = r_x[0]$

Convolution theorem (stochastic)

If $y[n] = h[n] * x[n]$, then: $P_y(\omega) = |H(\omega)|^2 P_x(\omega)$

11 A/D I - Interpolation & Sampling

The Continuous-Time World

CT vs. DT paradigm

D: $x_d[n] \in \ell_2(\mathbb{Z})$, $n \in \mathbb{Z}$, $\omega \in [-\pi, \pi]$, DTFT
A: $x_c(t) \in L_2(\mathbb{R})$, $t \in \mathbb{R}$ [s], $f \in \mathbb{R}$ [Hz], CTFT

Use cases

- $x(t) \rightarrow$ [processing] $\rightarrow y(t)$
- $x[n] \rightarrow$ [processing for synthesis] $\rightarrow y(t)$
- $x(t) \rightarrow$ [processing for analysis] $\rightarrow y[n]$

Real-world frequency

$f = \# \text{rep/s} = \text{Hz}$; $\Omega = 2\pi f$

CTFT (more info in cs)

- NOT PERIODIC
- NO LIMIT FREQUENCY

BandLimited (BL) functions

A CT signal is F_S -BL $\Leftrightarrow X(f)$ for $|f| > F_S/2$

Support of a function

The interval where the function is non-trivially zero

Sampling theorem (Shannon–Nyquist)

Any signal $x(t)$ whose highest frequency component is F_N Hz can be sampled with **no loss of information** using a sampling frequency:

$$F_s \geq 2F_N \quad (\text{i.e. } T_s \leq \frac{1}{2F_N})$$

Interpolation $x[n] \rightarrow x(t)$

Interpolation requirement

- decide on T_s (sampling period)
- make sure $x(nT_s) = x[n]$
- make sure $x(t)$ is smooth

Interpolations pros and cons

Polynomial: $+ C^\infty$ (max. smooth) – kernels $L_n^{(N)}$ depend on N

Local: $+ \text{kernel indep. of } N$ – lacks smoothness

Remarkable result

$$\lim_{N \rightarrow \infty} L_n^{(N)}(t) = \text{sinc}(t - n)$$

Sinc interpolation

$$x(t) = \sum_{n \in \mathbb{Z}} x[n] \text{sinc}(t - n)$$

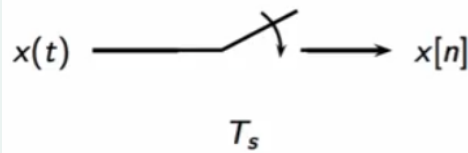
$\text{sinc}(m - n) = \delta[m - n]$, $\forall m, n \in \mathbb{Z}$
CTFT pair ($T_s = 1/F_s$):

$$\varphi(t) = \text{sinc}\left(\frac{t}{T_s}\right) \longleftrightarrow \Phi(f) = \frac{1}{F_s} \text{rect}\left(\frac{f}{F_s}\right)$$

12 A/D II - Aliasing

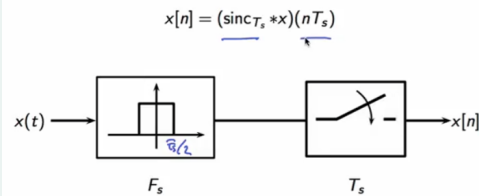
Raw Sampling

$$x[n] = x(nT_s)$$



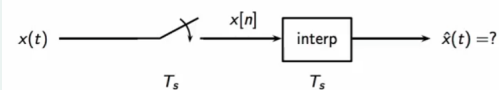
Sinc Sampling

$$x[n] = \left\langle \text{sinc}\left(\frac{t - nT_s}{T_s}\right), x(t) \right\rangle$$



Aliasing

Sinc Sampling



Map the frequency into $[-F_s/2 ; F_s/2]$ to make the signal F_s -BL.

13 A/D III - Multirate

Downsampling by N "DTFT pov"

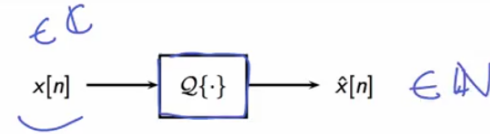
Copier à tout $2\pi/N$. Multiplier les indices par N .

Upsampling by N "DTFT pov"

Montrer la périodicité. Diviser les indices par N .

14 A/D IV - A/D and D/A conversion

Quantization schemes



Quantization MSE

$$\sigma_e^2 = E[(x - Q(x))^2]$$

Error analysis for uniform quantization

Error energy: $\sigma_e^2 = \Delta^2/12$, $\Delta = (B - A)/2^R$

Signal energy: $\sigma_x^2 = (B - A)^2/12$

SNR: $\text{SNR} = \sigma_x^2 / \sigma_e^2 = 2^{2R}$

SNR_{dB}: $10 \log_{10} 2^{2R} \approx 6R \text{ dB}$

A linear filter cannot improve the SNR due to quantization if the quantizer matches the input.